



PROFESSIONAL CERTIFICATION

MERN stack Web Development Training

Master the MERN stack MongoDB Express React Node.js to build modern full-stack web applications. Learn to create scalable dynamic and production-ready web solutions from frontend to backend.



60

Hours



Certified

Trainers



Industry

Expert



Professional Training | Industry-Recognized Certificate

Dubai, UAE

About Orbit Training Center

Your Gateway to Professional Excellence in Dubai



Who We Are

Established in 2023, Orbit Training Center is a leading professional training provider in Dubai, offering 60+ professional certification courses across diverse fields, including engineering, IT, design, and languages.



Our Benefits

-  Industry recognized certifications
-  Convenient location in the heart of Dubai
-  Affordable fees with multiple payment plans
-  Lifetime post-training support

★Our Features

-  Hands-on learning with real-world projects
-  Expert trainers with industry experience
-  Flexible learning options to fit your schedule
-  State-of-the-art infrastructure & facilities



Our Mission

We deliver outstanding courses, expert teaching, strong career guidance, and placement support to help every student achieve their best results. We stay connected with alumni through networking events, lifelong learning, and ongoing mentorship. We run our operations with smart systems and data-driven decisions to create a smooth, exceptional experience for all.

MERN stack Web Development

Build complete full-stack web applications using MongoDB Express React and Node.js.

Course Description

This comprehensive MERN stack course teaches you to develop end-to-end modern web applications. You will master MongoDB for database Express and Node.js for backend React for frontend state management RESTful APIs authentication JWT security and deployment. Through multiple hands-on projects build real-world full-stack applications that are fast

Duration

60

Hours of Training

Format

Classroom / Online

Flexible Option

Key Outcomes

- ✓ Build RESTful APIs with Node.js and Express
- ✓ Create NoSQL databases with MongoDB
- ✓ Develop dynamic user interfaces with React
- ✓ Manage application state with Redux
- ✓ Implement user authentication using JWT
- ✓ Connect frontend and backend seamlessly
- ✓ Deploy full MERN applications to cloud platforms

Target Audience

- full-stack developers learning modern technologies
- Frontend developers adding backend skills
- Students building complete web apps

Certification

- Certificate of Course Completion
- Industry Recognized Certification
- Trained by Certified Professionals

Prerequisites

- Basic knowledge of HTML CSS and JavaScript
- Understanding of ES6 features and modern JavaScript
- Familiarity with basic web development concepts

MERN stack Web Development

Build full-stack apps with MongoDB, Express, React, and Node.js from basics to deployment.

01 HTML Fundamentals

- > HTML document structure and DOCTYPE declaration
- > Headings paragraphs links and image tags
- > Lists forms and input elements
- > HTML5 semantic elements: header nav main section footer
- > Tables and embedding media with audio and video

02 CSS Fundamentals

- > Linking CSS to HTML and understanding the cascade
- > Selectors: element class ID and pseudo-classes
- > Box model: margin padding border and content
- > CSS Flexbox for building flexible layouts
- > Responsive design with media queries and mobile-first approach

03 JavaScript Essentials for MERN

- > ES6+ features: let const arrow functions and template literals
- > Destructuring spread and rest operators
- > Promises and async/await for asynchronous programming
- > Array and object methods for data manipulation
- > Modules: import and export syntax

04 Node.js Fundamentals

- > What is Node.js and how the event loop works
- > Node.js core modules: fs path and http
- > Creating a simple HTTP server with Node.js
- > npm: installing managing and using packages
- > Understanding package.json and node_modules

05 Introduction to Express.js

- > What is Express.js and setting up an Express server
- > Defining routes with GET POST PUT and DELETE methods
- > Using middleware in Express (built-in and third-party)
- > Handling request body with express.json and express.urlencoded
- > Serving static files and handling errors

06 Building RESTful APIs with Express

- > Understanding REST principles and HTTP status codes
- > Creating a full CRUD API with Express
- > Organizing routes using Express Router
- > Validating and sanitizing incoming request data
- > Sending structured JSON responses from the API

MERN stack Web Development

Build full-stack apps with MongoDB, Express, React, and Node.js from basics to deployment.

07 MongoDB Fundamentals

- > What is MongoDB and how NoSQL databases work
- > Installing MongoDB and using MongoDB Compass
- > Understanding collections documents and BSON format
- > Basic CRUD operations in the MongoDB shell
- > Connecting a Node.js application to MongoDB Atlas

08 Mongoose ODM

- > What is Mongoose and setting up schemas and models
- > Schema data types validators and default values
- > Creating reading updating and deleting documents with Mongoose
- > Understanding references and population between collections
- > Error handling in Mongoose operations

09 React Fundamentals

- > What is React and how the virtual DOM works
- > Setting up a React app with Create React App or Vite
- > JSX syntax and rendering components to the DOM
- > Props: passing and receiving data between components
- > State management with the useState hook

10 React Components and Hooks

- > Functional components and structuring a component tree
- > useEffect hook for side effects and data fetching
- > Handling forms and controlled components in React
- > Conditional rendering and rendering lists with map and keys
- > Building and reusing custom components across the app

11 React Router

- > Installing and setting up React Router
- > Defining routes with BrowserRouter Route and Routes
- > Navigating between pages with Link and NavLink
- > Dynamic routes using useParams for parameterized URLs
- > Programmatic navigation with useNavigate

12 Connecting React Frontend to Express Backend

- > Making HTTP requests from React using the Fetch API
- > Consuming REST API endpoints in React components
- > Handling loading states and error responses
- > Using environment variables to store the backend base URL
- > Understanding and resolving CORS issues

MERN stack Web Development

Build full-stack apps with MongoDB, Express, React, and Node.js from basics to deployment.

13 Authentication with JWT

- > User registration and password hashing with bcrypt
- > Generating and verifying JSON Web Tokens
- > Protecting backend routes with JWT middleware
- > Storing tokens on the frontend and sending with requests
- > Implementing login and logout flow in React

14 State Management with Context API

- > What is global state and when to use Context API
- > Creating a React Context and Provider
- > Consuming context values with the useContext hook
- > Managing authentication state globally across the app
- > Updating and sharing state across multiple components

15 Deployment

- > Preparing the Express backend for production
- > Deploying the Node.js backend to Render or Railway
- > Building the React frontend for production with npm run build
- > Deploying the React frontend to Vercel or Netlify
- > Connecting deployed frontend and backend with environment variables

16 MERN Capstone Project

- > Planning and designing a full-stack MERN application
- > Building the backend with Express MongoDB and Mongoose
- > Developing the React frontend with routing and state management
- > Implementing user authentication with JWT
- > Connecting and deploying the complete MERN application

Final Projects and Assessment:

- **Assessment:** Project-based evaluation through real-world projects to assess practical skills and application without theory exams.
- **Tailored Curriculum:** Customized to industry standards, covering essential topics for professional development in relevant fields.
- **Hands-on Projects:** Real-world simulations and interactive tasks to build applicable skills and create professional portfolios.
- **Certification Prep:** Targeted review sessions, practice exercises, and strategies for achieving industry certifications.
- **Ongoing Support:** Access to recorded materials, forums, instructor guidance, and career assistance for long-term growth.

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Muppuli Raja



One of great institute for REVIT BIM course. I completed my training with Mr Ahmed. He is one great trainer in this institute. He has very good knowledge for BIM Training. I completed my REVIT BIM training with orbit and it was very nice learning journey for me. And one of best thing is training center is very flexible for classes.



Arun



Best SolidWorks training in Dubai! The instructor made 3D modeling and assembly design easy to understand. The course covered all essential tools I need for product design and engineering work. Best training center for solidworks training. Best Training Center for Best Skills



Harshu CH



My experience at Orbit training center has been the best In every step I learn from there very clearly almost complete my course it's very interesting the trainer is the best professional and friendly ,and the rest of the team keep it up, thanks for supporting me Orbit and all staff...



Ranjan Axalan



I just finished my course in 3ds max. I learned a lot on how to create 3d model. Thank you for the experience and for the teaching method it help me to understand easily and grow my knowledge. I will use my learnings to my future company. Highly Recomanded!



Numan



I had good support from trainer Mr Ahmed and the other staff. They would schedule my BIM classes as per my convenience and would retake classes if I required. They always answer my calls when I need assistance. Not only that the trainer would go a mile a ahead by helping me with my work related issues like a true mentor.



Lelanie Fernandez



Overall, I highly recommend the Fusion 360 training at Orbit Center, especially with Ahmad as the instructor. This course has exceeded my expectations, and I am learning a lot. I would rate it 5 out of 5 stars!

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Aman Bakhtiyar



I had a great learning experience at Orbit Training Center. The trainers are knowledgeable, supportive, and always ready to help. The clear and practical teaching methods really helped improve my skills and confidence. I highly recommend Orbit Training Center to anyone looking to learn AutoCAD, Revit, and other professional skills.



Sana Zahid -



I recently completed the online Digital Marketing course with Orbit Training Center, and it was a great experience! The lessons were clear, practical, and easy to follow. The trainers were knowledgeable and supportive throughout the course. I've learned a lot that I can apply to my work here in Belgium. Highly recommend Orbit Training Center to anyone looking to improve their digital marketing skills!



Cleavon Henderson



Orbit Training Center offers outstanding CAD classes that are easy to understand, even as a first-year mechanical engineering student. The instructors simplify complex concepts and use practical examples, making the learning process engaging and effective. Whether you're new to CAD or improving your skills, Orbit Training Center provides a supportive and enriching environment. Highly recommended!



Danish Khan



Just finished Primavera P6 training at Orbit Training Center and it was fantastic! The trainer made complex scheduling concepts easy to understand with real-world examples. Got hands-on experience with project planning, resource allocation, and progress tracking. The course was well-structured and I feel confident using Primavera P6 for my projects now. Great value and would definitely recommend to other project managers!



500+ Students Trained

Join our growing community of professionals

95%

Satisfaction Rate

60+

Courses Offered

4.6

Average Rating



GET STARTED TODAY

Get in Touch with Us

At Orbit Training Center, we offer a diverse range of professional courses designed to equip you with the latest industry skills and knowledge. Our comprehensive programs cover in-demand fields such as Interior Design, Computer Science, Graphic Design, Digital Marketing, Solar Energy, and specialized software training including STAAD Pro, Revit Architecture, Tekla Structures, and more. Each course is crafted by industry experts to provide hands-on learning, real-world applications, and certification preparation.

Up to 30%

Early Bird Discount

Free

Demo Session



Call Us Now

+971-585450455

Contact Information



Visit Us

**211, Pinnacle Building
Sheikh Zayed Road, Al Barsha
Dubai, United Arab Emirates**



Phone

+971- 48857590



Email

training@orbittraining.ae



Website

www.orbittraining.ae



Working Hours

9:00 AM - 9:00 PM Monday- Sunday

